

Keegan S. McFadden

Virginia Tech — B.S. Computer Science (Spring 2026)

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TECHNICAL SKILLS

Languages: Java, C, Python

Concepts: Spring Boot, RESTful APIs, Data Structures & Algorithms, OOP, Caching, File I/O

Tools: Git, Linux, Docker, VS Code, Eclipse

PROJECTS

Procedural Terrain Generation Engine — *Java, Noise Algorithms*

- Built a **real-time 2D terrain engine** using **multi-octave fractal noise** and **deterministic hashing**
- Implemented **chunk-based world streaming**, smooth camera movement, and **biome logic**

Cybersecurity Simulation & Research Platform — *Flask, Python, SQLite, HTML*

- Worked on signalShift, a full-stack **web app** simulating phishing and social-engineering scenarios
- Developed tools for **scenario creation**, **advice configuration**, and **participant management**
- Added **automated data-collection** and **spreadsheet export features**. Deployed the platform on **AWS**

Outfit Recommendation Platform (Backend) — *Java, Spring Boot, REST APIs, Agile*

- Developed a **Spring Boot backend** to generate outfit recommendations from user wardrobe data
- Implemented a **weighted scoring algorithm** based on color matching, weather, and formality
- Designed **RESTful** endpoints to support frontend integration

Concert Tracker Application — *Swift, Xcode, REST APIs*

- Integrated the **Ticketmaster REST API** to enable concert search/saving by location and keyword
 - Implemented logic to **calculate total ticket prices** based on user selections
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COURSEWORK & ADDITIONAL EXPERIENCE

- Data Structures (Skip List, BST, AVL Tree, etc.) & Algorithms (quick sort, merge sort, etc.)
- Artificial Intelligence (N-grams, Neural Networks)
- Agile Software Development
- Mobile App Development (Swift)
- Clubs
 - CyberVT
 - Center for HCI Student Council @ VT
 - League of Legends Club @ VT